SDA ASSIGNMENT #1

Group Members

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Introduction

The SeniorHub platform is designed to help final year students efficiently manage their academic, project, and career-related responsibilities. In many university environments, final year processes like forming Final Year Project (FYP) groups, finding supervisors, applying for initial jobs, and getting graduation clearance are handled through scattered, manual, and inefficient channels.

This system provides a centralized digital platform where senior students can collaborate. Users can form FYP groups, browse faculty directories, send supervision requests, submit project deliverables, and track their graduation clearance checklists in real time. Additionally, the system features a career module for job applications and alumni networking.

The system improves transparency, communication, and efficiency in managing the chaotic final year experience. It ensures that students stay on top of critical deadlines, faculty can easily manage project groups, and administration can streamline the graduation clearance process.

Functional Requirements

FR ID	Priority	Description	Actor	Status
FR-01	High	The system shall allow users to register using university email.	All Users	Mandatory
FR-02	High	The system shall allow users to log in securely.	All Users	Mandatory

FR-03	High	The system shall allow students to create an academic/professional profile.	Student	Mandatory
FR-04	Medium	The system shall allow users to reset forgotten passwords.	All Users	Mandatory
FR-05	High	The system shall allow a student to create an FYP group and generate an invite link.	Student	Mandatory
FR-06	High	The system shall allow students to join an FYP group using an invite code.	Student	Mandatory
FR-07	Medium	The system shall provide a directory of faculty members and their research domains.	System	Mandatory
FR-08	High	The system shall allow group leaders to send FYP supervision requests.	Student	Mandatory
FR-09	High	The system shall allow groups to upload initial FYP proposal documents.	Student	Mandatory
FR-10	High	The system shall display upcoming FYP milestones and deadlines.	System	Mandatory
FR-11	High	The system shall allow students to submit final FYP deliverables (reports/code).	Student	Mandatory
FR-12	High	The system shall allow faculty to review and grade submitted deliverables.	Faculty	Mandatory

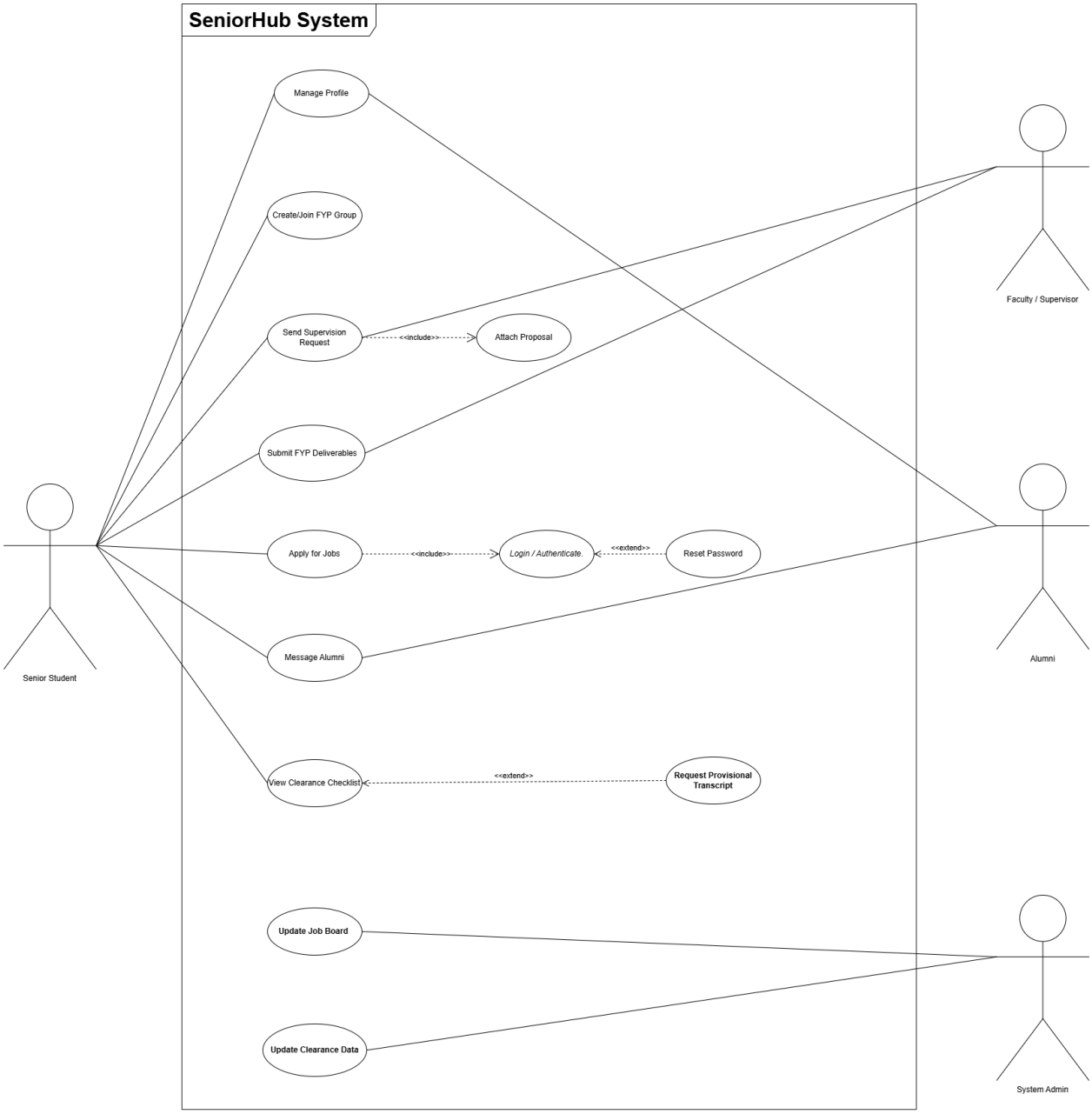
FR-13	Medium	The system shall display a job and internship board.	System	Mandatory
FR-14	High	The system shall allow students to apply for jobs using their profile resume.	Student	Mandatory
FR-15	Low	The system shall provide a searchable Alumni Directory.	System	Optional
FR-16	Medium	The system shall allow students to send direct messages to Alumni.	Student	Optional
FR-17	High	The system shall display a personalized graduation clearance checklist.	Student	Mandatory
FR-18	High	The system shall allow the Admin to update a student's clearance status.	Admin	Mandatory
FR-19	Medium	The system shall allow students to request a provisional transcript.	Student	Optional
FR-20	Medium	The system shall send automated notifications for deadlines and clearance updates.	System	Mandatory

Non-Functional Requirements

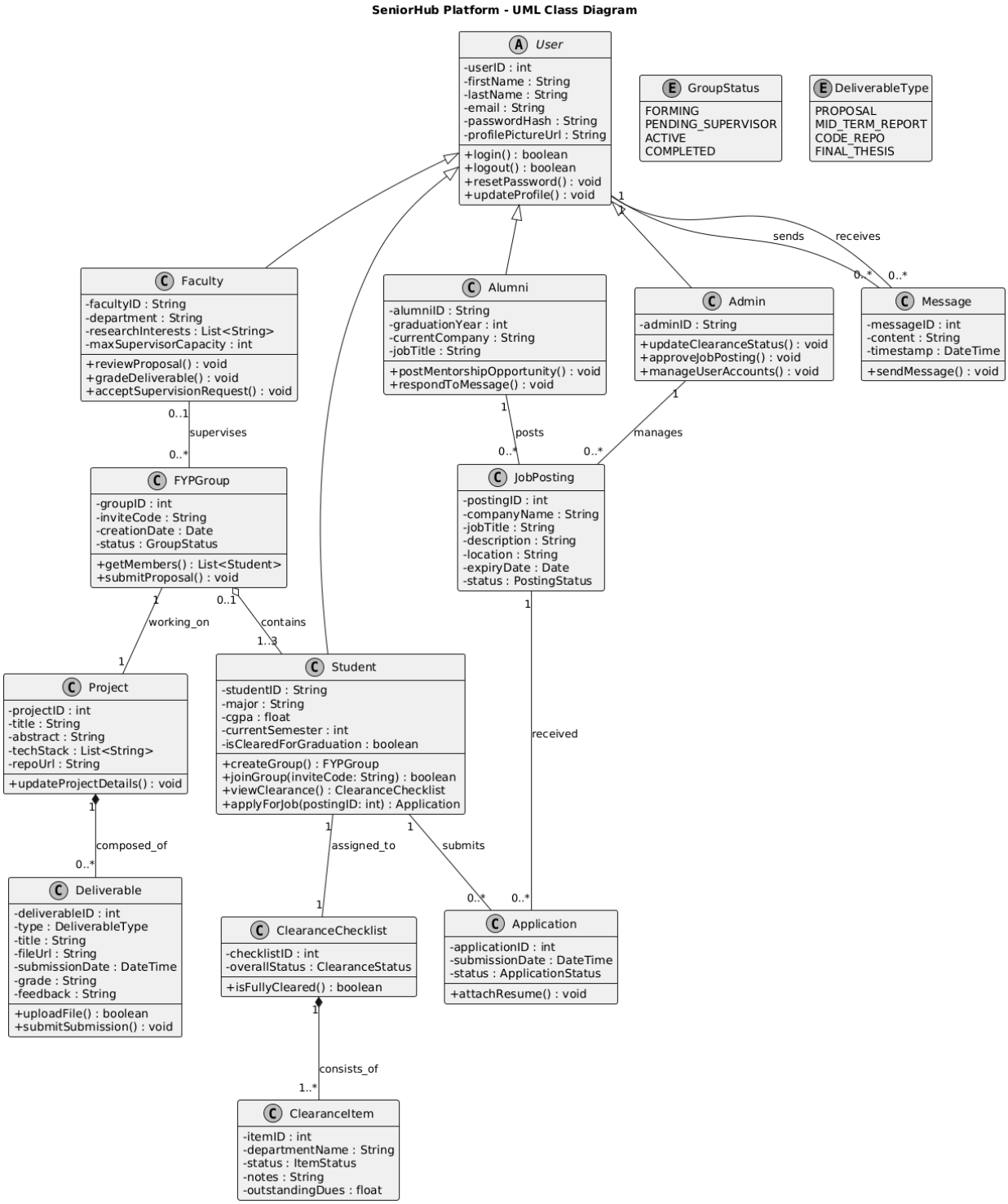
NFR ID	Category	Requirement Description
NFR-01	Availability	The system shall ensure an uptime of 99.9% for uninterrupted access.
NFR-02	Performance	The system shall load user dashboards and process requests in under 2 seconds.
NFR-03	Scalability	The system shall support at least 5,000 concurrent users without lag.
NFR-04	Security	The system shall hash all passwords and encrypt data using HTTPS/TLS.
NFR-05	Security	The system shall ensure data confidentiality and integrity.
NFR-06	Usability	The user interface shall be responsive and functional on both mobile and desktop.
NFR-07	Compatibility	The system shall be compatible with all major browsers (Chrome, Safari, Edge).
NFR-08	Reliability	The system shall perform automated daily backups of all student and FYP data.

NFR-09	Data Integrity	The system shall strictly prevent a student from joining multiple active FYP groups.
NFR-10	Access Control	The system shall strictly manage permissions based on user roles (Admin, Faculty, Student, Alumni).

Use-Case Diagram



Class Diagram



UML Specifications

Class	UML Specifications
Student	1. A Student can create or join one FYP Group. 2. Students can view their clearance checklist and apply to Job Postings. 3. Student accounts are managed by the System Admin. 4. Students can send messages to Alumni.
Faculty	1. Faculty can receive multiple Supervision Requests from different FYP Groups. 2. Faculty can accept or reject requests based on their availability. 3. Faculty reviews and updates the status of Deliverables submitted by their assigned groups.
FYP_Group	1. An FYP Group consists of 1 to 3 Students. 2. The group generates one unique invite code. 3. A group is supervised by exactly one Faculty member once a request is approved. 4. The group submits multiple Deliverables throughout the year.

Supervision_Request	1. Generated by the Student (Group Leader) and sent to the Faculty. 2. Contains an attached proposal document and a status (Pending, Accepted, Rejected).
Deliverable	1. A deliverable is submitted by an FYP_Group. 2. It contains file attachments (code/reports) and submission timestamps. 3. Must be linked to a specific project milestone.
Clearance_Record	1. Managed and updated by the System Admin. 2. Contains multiple departmental statuses (e.g., Finance, Library, Labs). 3. Can be viewed in real-time by the associated Student.
Job_Posting	1. Managed by the Admin or posted by Alumni. 2. Contains job descriptions, requirements, and application deadlines. 3. Receives applications (resumes/profiles) from Students.
Alumni	1. Alumni have profiles detailing their current company and job title.

	2. They can receive and respond to direct mentorship messages from Students.